

Research Project

The goal of this project is to use technical features/signals and machine learning methodologies to predict stock returns and construct a trading portfolio to evaluate the predictions.

Requirements:

- This is a take-home project, and you need to develop a trading signal based on real data, with python/pytorch or a data science suite of your choice.
- You will need to submit a PDF case study report, with a maximum of 10 pages. Code submission is not required.
- We expect you to spend no more than 3 days (part-time) on this project, which means you will need to send back the report via email no later than 3 days from the receipt of this project.

Data acquisition:

Download daily S&P 500 stock data from below link.

<https://www.kaggle.com/datasets/darkmatternet/s-and-p-500-stocks-25-years-of-data-updated-daily>

This link includes a sp500_stocks.csv file, which contains the complete daily stock OHLCV price data (Open, High, Low, Close, Adj Close, Volume) for all current S&P 500 companies from January 2000 to today — over 25 years of market history and 3+ million rows of data.

Above link also provides a sp500_companies.csv file which contains the sector and sub_industry information for all the companies.

Project guidelines:

1. In this project, we expect you to demonstrate data science skills on a real-life problem:
 - Gain an understanding of the dataset
 - Generate features/signals that can be useful to predict future returns
 - Prepare the dataset for model training
 - Based on your understanding of the dataset, choose an appropriate model and train model
 - Set up a workflow for model assessment and selection
2. Data and models
 - The data we provided (OHLCV data and sector/sub_industry data) is raw information about these stocks which can be used to construct features/signals.
 - Predictive features/signals that are constructed from raw information. Strong features are commonly referred to as alphas; Weaker features can be combined using statistical methods to generate stronger alphas. Some examples of the features/signals are: price momentum, Bollinger bands, MACD signals, etc.

- For the target, you can predict 1-day return or multiple-day return by your own choice. However, you could compare them or provide reasonings for your choice.
 - You can choose your own methodologies to train the data and combine features/signals, for example linear regression model, tree model, or neural network-based deep learning models, etc. Please provide your reasonings as well.
3. Validation and backtest
 - Please illustrate how you want to evaluate the quality of your predictions
 - You can also run a backtest/simulation to show the effectiveness of your predictions
 4. Important tips
 - We want to have one or multiple signals that have strong and consistent prediction power over a future horizon
 - Be disciplined with your in-sample, out-sample, focus on generalizability rather than in-sample result.
 - The prediction should focus on predicting relative return of stocks, rather than raw return of stocks. Raw return is extremely hard to predict, instead, predicting the relative performance of one stock against a sub-group or against the rest of the stocks. You can also think about the reason.

PDF submission guidelines:

The PDF submission would be able to demonstrate your data science skills. We expect you include below components in the report.

1. Feature engineering
2. Model training, selection, and evaluation
3. Your key algorithms or code structures
4. In each step, provide explanation as to why you made the choice
5. You can also consider discussing some of the topics below in the report:
 - overfitting
 - look-ahead bias
 - survivorship bias
 - data quality issues
 - turnover / market impact
 - regime dependence
 - economic intuitions
 - any other thoughts/findings you have